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**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks: 30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I K. Reshma (192565062) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Reshma

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Apply Backtracking Search to Assign Nurses.

Given,

Nurses: N_1, N_2, N_3

words: ICU, General, Emergency.

constraints

1. N_1 cannot be assigned to emergency.
2. N_2 must be assigned before N_3
(ICU < General < Emergency)
3. Only one nurse per ward.
4. N_3 cannot be assigned to ICU.
5. Every nurse must be assigned exactly one word.

Backtracking steps

step 1: Assign N_1

possible words:

→ ICU

→ General

→ Emergency (Not allowed)

choose $N_1 \rightarrow ICU$

current Assignment:

$N_1 = ICU$

Step 2: Assign N_2

Remaining words:

* General ✓

* Emergency ✓

choose $N_2 \rightarrow General$

current Assignment:

$N_1 = ICU$

$N_2 = General$

constraint check:

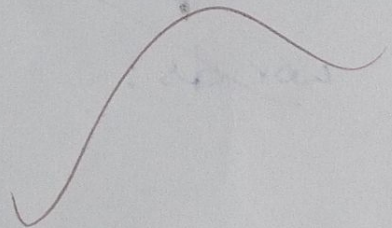
* one nurse per word ✓

* N_2 is before N_3 ✓

Step 3: Assign N_3

Remaining words:

* Emergency.



check constraints

→ N3 cannot be ICU ✓

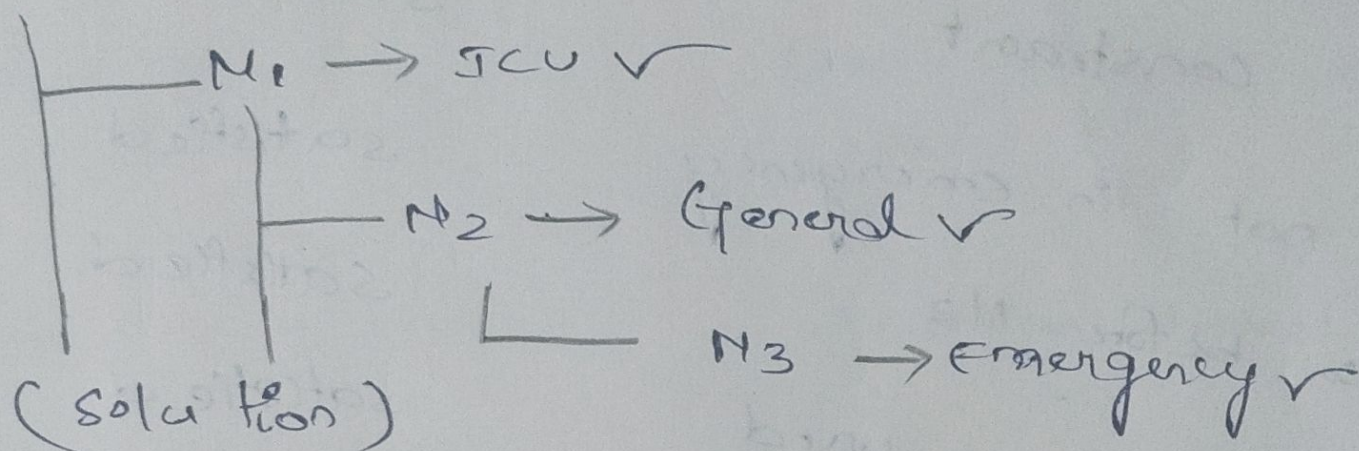
→ Emergency is acceptable ✓

→ N2 comes before N3 (Emergency) ✓

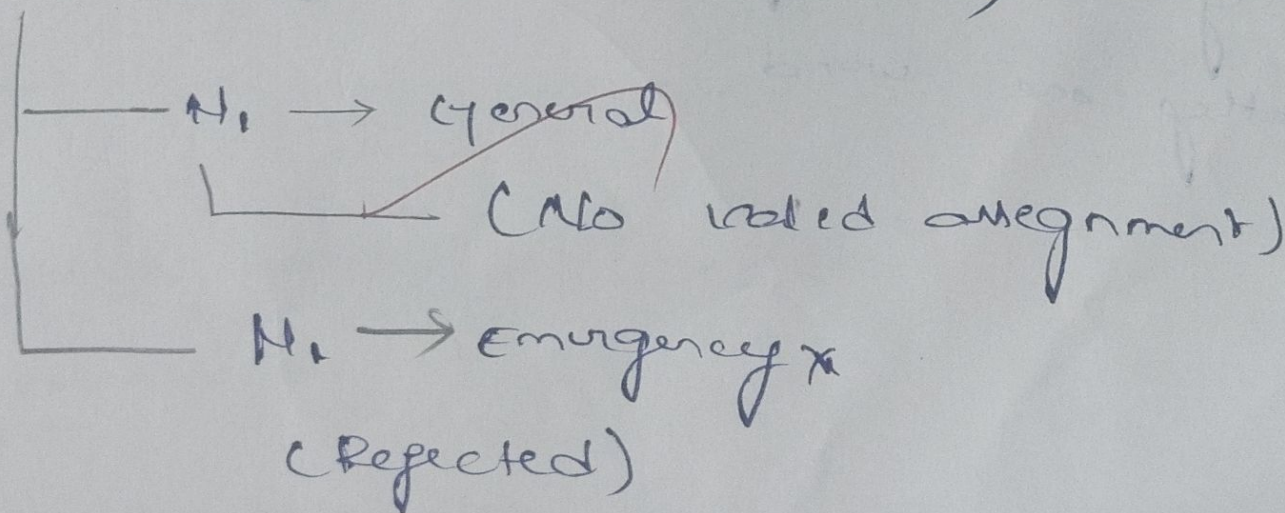
Assign: N3 → Emergency.

Backtracking Search Tree

start



N2 → Emergency x
(N3 must be General, but
N2 should be before N3)



final valid schedule

Nurse Assigned ward

N₁

ICU

N₂

General

N₃

Emergency

constraint verification

Constraint

status

N₁ not in Emergency

satisfied

N₂ before N₃

satisfied

one nurse per ward

satisfied

N₃ not in ICU

satisfied

Every nurse assigned

exactly one ward

satisfied.

Autonomous Rescue Drone

Constraint Analysis

constraint

Drone moves up, down, left, right

Blocked cells

limited sensing

Normal move cost = 1

Risky zone cost = 3

Dynamic environment

Effect on decision making

Movement limited to four directions.

Drone must avoid obstacles.

Drone observes only adjacent cells.

Reduces travel cost.

Drones avoid risky areas when possible.

Drone updates its map and replans its path.

(b) search strategy = uniform cost search

cost + online search

to start from the initial positions.

2. observe neighbouring cells.

3. Ignore blocked cells.

4. calculate movement cost:

* Normal cell = 1

* Risky cell = 3

5. Choose the path with the lowest total cost.

6. Move to the next cell.

7. Update the map if new obstacles are found.

8. Repeat until the survivor is reached.

Algorithm

Start

Initialize start position

cost = 0

while goal is not reached

observe neighbouring cells

remove blocked cells

calculate movement cost

Normal = 1

Risky = 3

choose minimum-cost path

Move to next cell

stop.

AI Concept map used

Intelligent Agent

- * It senses the environment using sensors.
- * It decides the best action automatically.

Rational Agent

- * select the safest and least-cost path.

Environment Type

- * Partially observable.
- * Dynamic
- * sequential
- * online search environment
- * uncertain.

(d) Path Reflection

uniform cost search is the best choice because

- * Finds the minimum-cost path.
- * Avoids risky zone whenever possible.
- * Handle blocked paths efficiently.
- * updates knowledge dynamically.
- * suitable for partially observable environments.

Advantages

- * optimal solution.
- * lowest travel cost.
- * Dynamic replanning.
- * safe navigation.
- * suitable for real-time disaster rescue.

Conclusion

The drone uses uniform cost search with an online search Agent to safely reach the survivor where minimizing travel cost and adapting to newly discovered obstacles.